DS 6021: Intro to Predictive Modeling
Quiz #2: October 2, 2026

Name: Quiz 2 Solutions

Quiz Format

This is a paper and pen/pencil quiz. It will be closed book. No reference materials of any kind will be allowed. You will have 20 minutes to answer the questions below. 1-3 sentences should be fine for each, and bullet points are also acceptable. The quiz is worth a total of **20 points**.

For this quiz, we are working with the following scenario: Consider a café that is modeling daily sales (in dollars) using two predictors: outdoor temperature (T) and a weekend indicator (W), where W=1 on weekends and W=0 on weekdays. The model for daily sales is represented by the following equation:

$$\widehat{sales} = b_0 + b_1 T + b_2 W + b_3 (TW)$$

A note on grading: Equivalent explanations and notation were accepted and partial credit was awarded for correct reasoning.

Question 1: Interpreting Interaction [6 points]

A manager at the café says, "b_1 tells us how sales change with temperature on any day of the week." Correct this statement by identifying the temperature slope on weekdays and weekends. What does b_3 represent?

Solution: The coefficient b_1 is the temperature slope for weekdays, when W = 0. The weekend temperature slope is $b_1 + b_3$. Thus, b3 is the weekend-minus-weekday difference in slopes: the predicted change in sales per one-degree increase in temperature is b_3 dollars greater on weekends.

Points:

- 2 points: Weekday slope is b_1.
- 2 points: Weekend slope is $b_1 + b_3$.
- 2 points: Interprets b_3 as the weekend-minus-weekday difference in temperature slopes. Award 1 of these 2 points for identifying a difference in slopes without specifying its direction.

Question 2: Comparing Groups [4 points]

This same manager then says, "Weekend sales are always predicted to be b_2 dollars higher than weekday sales at the same temperature." Is this correct? Explain.

Solution: Not generally. At temperature T, the predicted weekend-minus-weekday difference is:

$$\widehat{sales}_{weekend} - \widehat{sales}_{weekday} = b_2 + b_3 T$$

The coefficient b_2 alone describes the difference at T = 0. The difference is constant across temperatures only if $b_3 = 0$.

Points:

- 1 point: Rejects the claim in general.
- 2 points: Gives $b_2 + b_3T$ or clearly explains that the difference depends on temperature through the interaction. (The equation does not need to be included to get full credit if the explanation is correct.)
- 1 point: Explains that b2 is the difference at T = 0.

The special case b3 = 0 is useful but not required for full credit.

Question 3: Adding Predictors [5 points]

Another colleague at the café says that they are fairly certain that on days where the café offers specials on the menu, overall sales are boosted significantly and you should include this as part of your model. You add this term to the model and find that the R^2 value goes up, but the Adjusted R^2 goes down. Should you listen to your colleague and add this term to the model? Explain.

Solution: No, you should not listen to your colleague. R^2 always stays the same or goes up when you add predictors to a model, even if they are not meaningful. Adjusted R^2 adds a penalty for complexity (increased degrees of freedom), so it can decrease even when R^2 increases. This means that adding this term to the model does not justify the added complexity.

Points:

- 1 point: No, you should not listen to this colleague.
- 2 points: R^2 always stays the same or goes up when you add terms to the models even if they are not meaningful.
- 2 points: Adjusted R^2 penalizes additional predictors/degrees of freedom, and will decrease if the improvement to the model is insufficient.

Note: I'm not sure I properly emphasized that R^2 either stays the same or increases in class, so if you stated that R^2 always goes up, you will receive full credit, since that is my mistake for not properly covering this nuance. It can stay the same if the new coefficient is set to zero.

Question 4: Holdout sets and cross-validation [5 points]

As you look further into the model predicting sales, you learn that the person who built it did not use any kind of train/test split or cross-validation method. You decide to rebuild this model and take the following steps:

- Using the available data, you calculate the RMSE of the current model. (For this part there is no train/test split and you use all the data used to train the model in the RMSE calculation.)
- You rebuild the model using k-fold cross-validation (5 folds) with the same set of features (no feature selection), and get an RMSE that is higher than the original model but all coefficients are the same to within 95% confidence intervals.

Why should the k-fold cross validation give you more confidence in the predictions made by this model, even though the RMSE is higher?

Solution: The original RMSE was calculated on the same data used to fit the model, because it is measuring how well the model did on data it has already seen, so we expect it to be lower. The k-fold cross validation calculates the RMSE score on data it has never seen, and it does this over multiple folds, so it gives a more honest assessment of out of sample error. Because the coefficients stay the same within 95% confidence intervals, this means that the original model is reasonably stable, and not just fitting noise.

Points:

- 2 points: Original RMSE is overly optimistic because it is evaluated on the same data that was used to train the model.
- 2 points: CV RMSE estimates performance on unseen data, so it is a more honest measure of predictive accuracy. A higher value is expected when we calculate the RMSE on out of sample data.
- 1 point: Coefficients that are consistent across the k-fold CV indicate that the model is stable and generalizes and we can trust the predictions with a reasonable degree of confidence.